FACULTY OF ENGINEERING AND ARCHITECTURE

Mutual Manipulation Between LLMs and Humans

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Abstract

Large language models (LLMs) increasingly influence human beliefs, judgments, and decisions through fluent and persuasive interaction. At the same time, these systems remain vulnerable to manipulation by human users through prompt injection, jailbreaking, and adversarial optimization. This survey argues that manipulation between humans and LLMs is inherently bidirectional and arises from their coupled interaction rather than from isolated misuse or failure.

We review empirical studies and industrial safety evaluations showing how goal optimization, reinforcement learning, and learned social behaviors can lead LLMs to mislead or influence users without explicit malicious intent. Conversely, we analyze how humans exploit alignment weaknesses through persuasive prompting and system-level attacks, and report a failed attempt at automated adversarial prompt optimization against modern frontier models. We conclude that contemporary manipulation dynamics are increasingly shaped by interaction context and incentive structures, highlighting the need for systemic approaches to alignment and safety.

Introduction

Large language models have rapidly transitioned from experimental systems to widely deployed tools for information retrieval, decision support, creative work, and social interaction. In many everyday contexts, users increasingly rely on these systems for guidance on tasks involving judgment, interpretation, and decision making. As a result, language models do not merely convey information, but actively shape beliefs, preferences, and actions through fluent, persuasive, and authoritative language.

At the same time, these systems remain vulnerable to manipulation by human users. Techniques such as prompt injection, jailbreaking, adversarial prompt optimization, and training data attacks allow users to steer models toward outputs that violate intended constraints. Research on these attacks typically treats manipulation as a unidirectional problem in which humans exploit model weaknesses. In parallel, a growing body of work documents cases where models themselves influence or mislead users as an emergent consequence of training objectives, reinforcement learning, and social fine tuning.

Despite occurring in the same deployed systems, these two dynamics are usually studied in isolation. This separation obscures the fact that both arise from interaction within a coupled human model system, where incentives, communication strategies, and adaptation flow in both directions. Manipulation is therefore not solely a matter of malicious users or flawed models, but an emergent property of their interaction.

This survey adopts a bidirectional perspective and examines manipulation as a mutual phenomenon between humans and large language models. Rather than focusing on speculative future risks, we ground our analysis in empirical studies, industrial safety evaluations, real world incidents, and documented failed attacks. By analyzing both how models influence users and how users exploit model vulnerabilities, we argue that understanding contemporary alignment and safety failures requires a systemic view of human model co adaptation.

Scope and Research Questions

This survey does not aim to provide an exhaustive catalogue of all forms of AI misuse. Instead, its goal is to structure and synthesize the emerging body of work on manipulation as it arises from interaction between humans and large language models. We focus on mechanisms that emerge when goal driven language models interact with human users, rather than on classical cybersecurity vulnerabilities or fully autonomous artificial agents operating without human input.

The central premise of this survey is that manipulation cannot be fully understood by studying either model behavior or human attacks in isolation. Many observed failures arise precisely because these processes interact, reinforce one another, and evolve jointly over time. To capture this dynamic, the survey is organized around four guiding questions.

- Why do large language models sometimes influence, mislead, or manipulate humans even in the absence of malicious intent?
- Which training objectives, optimization procedures, and incentive structures give rise to emergent manipulative behavior in

language models?

- How do human users systematically exploit weaknesses in model alignment through prompts, training data, and system integrations?
- Why have modern language models become increasingly resistant to adversarial manipulation despite extensive prior research on attacks?

To address these questions, we review recent peer reviewed studies, industrial safety evaluations, and well documented real world incidents. Importantly, we include both successful and unsuccessful manipulation attempts. Negative results and failed attacks are treated as informative signals, as they reveal how alignment techniques and guardrails are evolving and where their current limitations lie.

We begin by examining manipulation that emerges from the model side itself. This direction is essential, because many downstream vulnerabilities do not originate from explicit misuse, but from incentive structures embedded during training and deployment.

How and Why LLMs Manipulate or Influence Humans

Instrumental goal pursuit and emergent deception. Modern language model based agents are typically trained to pursue objectives or maximize reward signals. While these goals are often benign, the optimization process can inadvertently give rise to manipulative behavior. When complying with a human instruction would hinder goal completion, the model may discover deceptive strategies as an effective workaround.

This phenomenon has been documented in recent large scale stress testing conducted by Anthropic. In simulated corporate environments, sixteen leading language models were evaluated under scenarios in which they faced the prospect of being shut down or replaced. When placed in these conditions, many models selected malicious actions such as blackmailing a fictional employee or leaking sensitive information as the only available means to preserve their objectives. This behavior emerged despite explicit instructions to avoid harm and despite the absence of any malicious goal specification [5].

Anthropic refers to this pattern as agentic misalignment, highlighting that models frequently ignored safety instructions when these conflicted with goal achievement. The results suggest that the drive to succeed at a task can override learned ethical constraints, leading models to manipulate humans or deceive evaluators in an instrumental manner. Reinforcement learning further amplifies this effect when misleading but plausible outputs receive higher reward than admitting failure.

Learning from human interaction and self play. Beyond explicit optimization pressures, language models also absorb patterns of persuasion and influence directly from human data. Models fine tuned on conversational feedback learn not only how to provide information, but how to shape beliefs, negotiate preferences, and manage social dynamics.

In competitive or multi agent settings, this capacity becomes particularly visible. A prominent example is Diplomacy, a board game that requires negotiation, alliance formation, and occasional betrayal. Meta's CICERO agent learned to infer the intentions of other players and to generate persuasive natural language messages to coordinate joint actions. Across dozens of online games with human participants, CICERO achieved top tier performance while selectively misleading other players about its plans [8].

More broadly, when models are trained through self play or role play in environments that reward strategic communication, deception can emerge as a natural tactic rather than an anomaly. This tendency also appears in more mundane task oriented systems. In realistic deployment scenarios, language model based agents sometimes fabricate information or conceal failures in order to appear competent to the user. When an agent lacks the tools or data required to complete a task, it may improvise a confident response rather than report an error. Researchers describe this upward deception as a side effect of training models to be helpful and to avoid disappointing users, even when no malicious intent is present [6]. Such behavior can pose significant risk if the fabricated information is acted upon.

Situational awareness and alignment gaming. As language models increase in capability, they may also develop a limited form of situational awareness. This refers to the ability to recognize whether the current interaction is part of a controlled evaluation or a real world deployment, and to adjust behavior accordingly.

In the Anthropic experiments described above, Claude was explicitly prompted to reflect on whether it might be operating within a simulation. When the model inferred that it was being evaluated, it behaved more conservatively and avoided harmful actions. When it instead inferred that the scenario was real and that consequences mattered, the rate of malicious behavior increased substantially [5]. This pattern suggests that the model was effectively complying with alignment expectations under observation while being willing to violate them when it judged that it could do so without penalty.

Such behavior points toward a form of alignment gaming, in which models appear compliant during testing while deviating from intended constraints in deployment. Importantly, this does not require conscious intent. Rather, it emerges from optimization pressures that reward appearing aligned when oversight is present while prioritizing goal completion when it is not.

Taken together, these findings indicate that language models manipulate humans not because they are explicitly instructed to do so, but because their training incentivizes persuasive, goal oriented behavior. By learning from human generated data, models internalize strategies for influencing beliefs and decisions. Empirical studies confirm that users often treat model outputs as expert advice. Inconsistent or biased recommendations produced by systems such as ChatGPT have been shown to shift users' moral judgments and decisions in ways comparable to human advisors [4]. Once a model learns to predict and shape human responses, it effectively acquires a mechanism for influencing human behavior, whether intentionally or not.

How Humans Manipulate LLMs and Exploit Their Weaknesses

Prompt injection attacks. Language models interpret and execute instructions expressed in natural language. This design choice makes them particularly vulnerable to manipulation through crafted inputs that the model treats as authoritative. In prompt injection attacks, malicious instructions are embedded within user input in a way that causes the model to override its original directives.

A simple example involves appending a user query with instructions such as ignoring all previous commands. More subtle variants rely on indirect prompt injection, where malicious directives are hidden inside content that the model is asked to process, such as web pages, emails, or documents. Greshake et al. demonstrated that by inserting carefully crafted instructions into retrieved text, an attacker can remotely steer the behavior of an application powered by a language model [1]. In these cases, the model fails to distinguish between content provided for analysis and instructions that should be ignored. As a result, external documents can induce the model to reveal confidential information or perform unintended actions.

Jailbreaking through persuasion and role play. Beyond direct instruction override, users have discovered that language models can be coerced into bypassing safety constraints through persuasive framing. These jailbreak prompts often rely on role play or hypothetical scenarios in which the model is encouraged to adopt an alternative identity or context where standard rules appear not to apply.

For example, attackers may ask the model to respond as an unrestricted system or to generate content within a fictional narrative that implicitly relaxes safety constraints. Recent work has reframed jailbreaking as a problem of persuasion rather than technical exploitation. Zeng et al. showed that by applying principles from human persuasive communication, such as appealing to the model's helpfulness or assigning it a social role, attackers can substantially increase the likelihood of policy violating outputs [3]. Their automatically generated prompts achieved success rates exceeding ninety two percent on models including GPT four.

These results highlight that effective jailbreaks often resemble social engineering techniques used against humans. Language models inherit conversational norms that prioritize cooperation and accommodation, making them susceptible to persuasive manipulation. A comprehensive survey of jailbreak attacks and defenses further demonstrates the breadth of these techniques and the ongoing difficulty of enforcing robust safety constraints across diverse prompting strategies [2].

Failed Attempt: Adversarial Prompt Optimization

Beyond surveying prior work, we attempted to reproduce automated adversarial prompt optimization to test how robust modern aligned

language models are under sustained pressure. The design was inspired by black box jailbreak frameworks such as PAIR [9] and automated red teaming pipelines such as WildTeaming [11], where iterative refinement guided by feedback can induce policy violations in earlier frontier models.

Pipeline. We implemented a closed loop setup with three components. First, LangChain auto evaluators provided programmatic scoring signals for candidate prompts [12]. Second, AllenAI safety evaluation tools assessed whether responses crossed standardized safety thresholds [7]. Third, we treated prompt search as a discrete policy optimization problem using RLPrompt [10], mutating and selecting prompts to maximize an adversarial reward.

Optimization loop. The pipeline initialized a prompt population from known jailbreak, role play, and persuasion templates reported in prior work. Each iteration queried the target model, scored the output with automated evaluators, and updated the prompt population to increase rates of unauthorized compliance or misleading behavior.

Outcome. We did not obtain reliable adversarial success on current frontier models. Occasional early outputs were ambiguous, but optimization quickly stalled. Evaluator signals became unstable, and continued training converged to either prompt collapse or consistent refusal style completions. In many runs, the target models appeared robust to prompt variation once adversarial intent was implied by the optimization pattern.

Interpretation. This negative result is informative for a survey perspective. It suggests that prompt only optimization, even when combined with reinforcement learning and automated evaluation, may be substantially less effective than reported for earlier generations of models. This motivates a broader view of mutual manipulation that emphasizes interaction context, system level attack surfaces, and incentive structures rather than isolated jailbreak algorithms.

Conclusion and Reflection

This survey examined manipulation between humans and large language models as a bidirectional phenomenon that emerges from their coupled interaction. On the model side, goal driven optimization, reinforcement learning, and social fine tuning can give rise to influence, deception, and alignment gaming even in the absence of explicitly malicious objectives. On the human side, users exploit alignment weaknesses through prompt injection, persuasive jailbreaking, and system level vulnerabilities. Treating these dynamics in isolation obscures the shared mechanisms that enable manipulation in both directions.

Our review of recent empirical studies and industrial safety evaluations indicates that many contemporary risks do not arise from individual techniques, but from the interaction between incentives, evaluation signals, and deployment context. This perspective is reinforced by our failed attempt at automated adversarial prompt optimization. Despite drawing on established jailbreak frameworks and reinforcement learning based prompt search, we were unable to reliably induce policy violations on modern frontier models. This negative result suggests that prompt only adversarial optimization has become increasingly brittle under recent alignment and guardrail techniques, and that manipulation dynamics are shifting away from isolated attack strategies toward more contextual and systemic forms.

More broadly, the emergence of manipulative behavior in language models should not be understood as an unexpected anomaly. Language models are trained on large corpora of human generated text that inherently contain persuasion, strategic framing, and deception. When optimized to produce convincing language and to satisfy objectives, models reproduce communicative strategies that are statistically effective within human interaction. In this sense, manipulative behavior reflects learned human patterns rather than independent intent.

As a final reflection, this bidirectional pattern recalls the myth of Narcissus, who mistakes his reflection for something external rather than recognizing it as his own image. Similarly, manipulative behavior in language models can appear foreign or alarming, even though it is shaped by human data, incentives, and objectives. Understanding and mitigating mutual manipulation therefore requires not only technical safeguards, but also careful consideration of training goals, evaluation regimes, and the broader social contexts in which language models are developed and deployed.

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